

13.3 Determining Points on a Circle

Lesson Introduction

 Benchmarks	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.			 Learning Targets	- I can determine whether a point is on the circle, in a circle, or outside a circle. - I can find the center of a circle when given two points. - I can write the equation of a circle given the endpoints of a diameter.	
 Benchmark Coherence	Building On		Working Toward			
	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.		N/A			
 Essential Questions	- How can you find the center of a circle when given two points? - What is the relationship between the radius and where a point lies within a circle? - How can you determine whether a point lies on the circle, on the inside, or on the outside of the circle? - How do you write the equation of a circle given the endpoints of the diameter?					
 Lesson Narrative	The lesson guides students on how to determine whether a given point is on the circle, inside the circle, or outside of the circle. Students are also expected to be able to write equations of circles and graph circles on the coordinate plane, referring to Unit 10. Students will also discover how to find the center of the circle when given two endpoints, and how to write the equation of a circle given the endpoints of the diameter.					
 Component Summary	13.3.1 Warm-Up (~5 min.)	Notice and Wonder (SE p. 294)	13.3.3 Try It (20 min.)	Practice on finding points on a circle (SE p. 298)		
	13.3.2 Exploration (20 min.)	Discovery on determining if a point lies inside, outside, or on a circle (SE p. 295)	13.3.4 Wrap-Up (~5 Min)	Ticket Out the Door (SE p. 299)		
 Resources	Glossary		Materials		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Graphing utility		If allowing students to use a graphing utility, prepare this before class starts.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with simplifying equations to determine if the simplified equation is equal to, greater than, or less than the radius squared. - Students may struggle with using the distance formula to find the diameter in 13.2.2, on question 2a.	Consider providing graph paper, or allow students to use a graphing utility, to help visualize circles that are not graphed in the lesson, and/or that have radii that are not whole numbers.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder Lesson 13.3 has multiple opportunities for students to notice and wonder. 13.3.1 has a Notice and Wonder to start the lesson, and 13.3.2 gives multiple opportunities for students to notice and wonder to create the conjectures that students are expected to form.	MA.K12.MTR.1.1: Participate in Effortful Learning Lesson 13.3 requires that students work both individually, and with other students. MTR 1 states that students should support one another and ask questions when working on a mathematical task. Students are expected to ask Notice and Wonder questions throughout the lesson, tying in the Literacy and Language Routine.
 Differentiate Instruction	Consider allowing students to use a graphing utility for much of this lesson. Using a graphing utility can help students better determine the relationship between the endpoints of a circle and the center. Using a graphing utility can also help students determine where a point lies on the circle, inside the circle, and outside the circle.	
Lesson Notes		

13.3.1 Warm-Up: Notice and Wonder

Component Overview: Students read a prompt about the picture and make observations on things they notice and wonder.



13.3.1 Warm-Up: Notice and Wonder

Castlerigg Stone Circle is one of the earliest stone circles to be found in Britain and is important in terms of megalithic astronomy and geometry. It is composed of 38 free standing stones, some measuring up to 10 feet high and dating back to the Neolithic period, 4000 to 5000 years ago.



1. What do you notice?

Answers will vary. I notice there are no other rocks except the ones making the circle.

2. What do you wonder?

Answers will vary. I wonder how they moved them and placed them in this location.



Consider allowing students to share their Notice and Wonder with their classmates, if time allows, as this is related to the Literacy and Language Instructional Routine for this lesson.

13.3.2 Exploration: Points on a Circle

Component Overview: This Exploration guides students through determining if coordinate points are on a circle or are on the inside or outside of a given circle. Students will make conjectures about the relationships between the equation of a circle and the location of a point. Students will use this distance formula to find the diameter, and the midpoint formula to find the radius.



13.3.2 Exploration: Points on a Circle

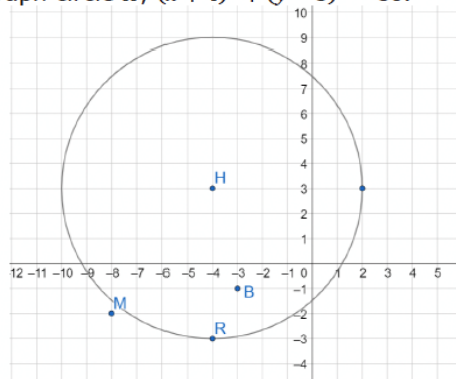
1. The equation of circle H is $(x + 4)^2 + (y - 3)^2 = 36$.

- a. Complete the table by substituting the given coordinates for x and y in the equation of circle H . Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$M(-8, -2)$	$(-8 + 4)^2 + (-2 - 3)^2 = 36$ $41 \neq 36$	$41 = 36$
$B(-3, -1)$	$(-3 + 4)^2 + (-1 - 3)^2 = 36$ $17 \neq 36$	$17 = 36$
$R(-4, -3)$	$(-4 + 4)^2 + (-3 - 3)^2 = 36$ $36 = 36$	$36 = 36$

- b. What do you notice about the simplified equations?
Answers will vary.
Two of the simplified equations are false statements and one is a true statement.

c. Graph circle H , $(x + 4)^2 + (y - 3)^2 = 36$.



This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group before releasing students to do 13.3.3 Try It. While circulating, look for students who may need assistance when substituting the given coordinate for x and y . Consider modeling the first coordinate for these students.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

13.3.2 Exploration: Points on a Circle (continued)

- d. Plot points $M(-8, -2)$, $B(-3, -1)$, and $R(-4, -3)$.

See graph.

- e. What do you notice about the relationship between the location of the points and their simplified equations?

Answers will vary. If the equation simplifies to the radius squared, then the point is on the circumference of the circle. If it simplifies to a number less than the radius squared, it's inside the circle. If it simplifies to a number greater than the radius squared, then it's outside the circle.

- f. Write a conjecture about the location of a point and the equation of the circle.

Answers will vary. If the coordinates are substituted into the equation and the equation is simplified to a true statement, then the point lies on the circumference of the circle.

- g. Ask two classmates for their conjectures. Record their conjectures and write your summary of their conjecture. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Conjecture	Initials
<i>Answers will vary.</i>		

- h. Review your conjecture with your partner. If necessary, make any changes.



For a visual approach, have students make a table with the conjectures. See the table below. This may help students organize their conjectures.

Radius Squared	Location of Point
$>$ Radius squared	Outside the circle
$<$ Radius squared	Inside the circle
$=$ Radius squared	On the circle



Give students time to discuss the conjectures, as this is the guiding point of the lesson. Allowing time for students to develop the conjectures will also tie to MTR 1.

13.3.2 Exploration: Points on a Circle (continued)

2. The equation of circle K is $(x + 6)^2 + (y + 5)^2 = 25$ with points $N(-6, 0)$, $V(-3, -1)$, $T(-1, -5)$, and $R(-9, -9)$ on the circle.

- a. What is the length of the diameter of circle K ?

$$r = \sqrt{(-6 + 6)^2 + (-5 + 0)^2} = 5; \quad d = 10$$

- b. Complete the table by finding the distance between the given points.

Points	Work	Distance
$N(-6, 0)$ and $R(-9, -9)$	$d = \sqrt{(-6 + 9)^2 + (0 + 9)^2}$	$\sqrt{90} \approx 9.487$
$V(-3, -1)$ and $R(-9, -9)$	$d = \sqrt{(-3 + 9)^2 + (-1 + 9)^2}$	$\sqrt{100} = 10$
$T(-1, -5)$ and $R(-9, -9)$	$d = \sqrt{(-1 + 9)^2 + (-5 + 9)^2}$	$\sqrt{80} \approx 8.944$

- c. What do you notice about the distances between the points?

Answers will vary. Only one point has a distance from point R that is equal to the length of the diameter.

- d. What can you conclude about points V and R ?

Points V and R are endpoints of a diameter of the circle.



Some students may struggle with finding the diameter of the circle. To help students, have the students find the radius using the equation, and then multiply by 2.



Consider employing a jigsaw routine with question 2b and assigning students one of the three problems to complete. Then, have students share the results of their problems with the class.



While circulating, ask students to share their notice and conclusion with you.

13.3.2 Exploration: Points on a Circle (continued)

3. Circle W has a radius of $\sqrt{29}$ and points $S(2, -2)$, $D(5, 1)$, and $H(12, -6)$.

- a. Determine which two points are the endpoints of a diameter of circle W . Show your work.

$$\text{Diameter} = 2(\sqrt{29})$$

$$SH = \sqrt{(2 - 12)^2 + (-2 + 6)^2} = \sqrt{116} = 2\sqrt{29}$$

Points S and H are the endpoints of a diameter of circle W .

- b. Describe how to find the center of the circle using the two endpoints of the diameter.

Use the midpoint formula to determine the midpoint of the diameter

- c. What is the center of the circle?

$$\left(\frac{2 + 12}{2}, \frac{-2 + 6}{2}\right) = (7, 2)$$

- d. Write the equation for circle W .

$$(x - 7)^2 + (y - 2)^2 = 29$$

4. The points $G(3, -1)$ and $H(3, 3)$ are the endpoints of a diameter of circle M . Write the equation of circle M .

$$\left(\frac{3 + 3}{2}, \frac{-1 + 3}{2}\right) = (3, 1)$$

$$GH = \sqrt{(3 - 3)^2 + (-1 - 3)^2} = 4; r = 2$$

$$(x - 3)^2 + (y - 1)^2 = 4$$



Consider asking partner pairs the following question to guide students through question 3:

- How can you determine which two points are the endpoints of the diameter?



Consider allowing students to use graph paper or an online graphing utility as a visual for question 4.



Consider asking partner pairs the following question to guide students through question 4:

- How can you determine the coordinates of the radius?
- How do you find the length of the radius when given the coordinate?
- What two pieces of information are needed to write the equation of a circle?

13.3.3 Try It: Points on a Circle

Component Overview: Students are working on skills that were completed during the 13.3.2 Exploration. Students should work on the questions individually.



13.3.3 Try It: Points on a Circle

1. The equation of circle P is $(x - 8)^2 + (y - 2)^2 = 13$.

Determine whether the following points lie inside, outside, or on circle P .

Point	Work	Location
(5, -1)	$(5 - 8)^2 + (-1 - 2)^2 = 18$ $18 > 13$	○ Inside circle P ○ Outside circle P ○ On circle P
(6,5)	$(6 - 8)^2 + (5 - 2)^2 = 13$ $13 = 13$	○ Inside circle P ○ Outside circle P ○ On circle P
(12,2)	$(12 - 8)^2 + (2 - 2)^2 = 16$ $16 > 13$	○ Inside circle P ○ Outside circle P ○ On circle P
(11,0)	$(11 - 8)^2 + (0 - 2)^2 = 13$ $13 = 13$	○ Inside circle P ○ Outside circle P ○ On circle P
(9,4)	$(9 - 8)^2 + (4 - 2)^2 = 5$ $5 < 13$	○ Inside circle P ○ Outside circle P ○ On circle P



Before releasing students to complete the questions, consider debriefing student questions, and addressing misconceptions you noted in 13.3.2 Exploration. Also, consider highlighting student voices to share rich discussions or comments overhead while they collaborated.



If students need further support, consider having students work with a partner or small group using the Instructional Routine, "Take Turns". Each student will take a turn in identifying the location of each point. The other student(s) will agree or disagree with the work, and can use this time to ask clarifying questions.

13.3.3 Try It: Points on a Circle (continued)

2. Circle T and circle R intersect at point H .

Equation of circle T is $x^2 + (y - 7)^2 = 20$

Equation of circle R is $(x - 7)^2 + (y - 3)^2 = 25$

- a. Which coordinate pair is the location of point H ?

- $(-4, 5)$
- $(7, 8)$
- $(2, 3)$
- $(4, 9)$

$$(2)^2 + (3 - 7)^2 = 20$$

$$(2 - 7)^2 + (3 - 3)^2 = 25$$

3. Circle S has a radius of 10 and points $F(-6, -6)$, $C(8, 8)$, $W(10, 2)$, and $M(-8, -4)$ on the circle.

- a. Determine which two of the given points are endpoints of a diameter of circle S .

$$\text{Diameter} = 20$$

$$CM = \sqrt{(-8 - 8)^2 + (-4 - 8)^2} = 20$$

$\overline{CM}$ is a diameter of Circle T

- b. Find the center of circle S .

$$\left(\frac{8 - 8}{2}, \frac{8 - 4}{2} \right) = (0, 2)$$

- c. Write the equation of circle S .

$$x^2 + (y - 2)^2 = 100$$



- What is the relationship between the location of the point and the equation of the circle?
- How can you determine if the point is on the circumference of a circle?
- How can you determine the point of intersection of the two circles?



Consider allowing students to use a graphing utility for questions 2 and 3 to provide as a visual.

13.3.4 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a question to determine mastery of the lesson.



13.3.4 Wrap-Up: Ticket Out the Door

1. The equation of circle W is $x^2 + (y + 4)^2 = 26$.

Select all the points that lie on circle W .

- ☐ $(3,0)$
- ☒ $(5,-3)$ $(5)^2 + (-3 + 4)^2 = 26$
- ☒ $(1,1)$ $(1)^2 + (1 + 4)^2 = 26$
- ☐ $(-3,-2)$
- ☒ $(-5,-5)$ $(5)^2 + (-5 + 4)^2 = 26$



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.



For students who finish early, have students create an equation of a circle, and three coordinate points that lie on the circle.